

Book Store Management System

Database Design Document

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DBMS – Week 2

Project Title: Online Book Store – An E-Commerce Platform with Reading Tracker

PURPOSE: Identification of Entities, Attributes, Primary Keys, and Relationships.

1. Entities and Attributes

Customer

Primary Key: Customer_ID

Attributes:

- Customer_ID (PK)
- First_Name
- Last_Name
- Email
- Phone_Number
- Password
- Address
- City
- State
- Postal_Code
- Registration_Date

Category

Primary Key: Category_ID

Attributes:

- Category_ID (PK)
- Category_Name
- Category_Type

Book

Primary Key: Book_ID

Foreign Key: Category_ID

Attributes:

- Book_ID (PK)
- Category_ID (FK)
- Title
- Author
- ISBN
- Description
- Price
- Stock_Quantity
- Image_URL
- Availability_Status

Order

Primary Key: Order_ID

Foreign Key: Customer_ID

Attributes:

- Order_ID (PK)
- Customer_ID (FK)
- Order_Date
- Total_Amount
- Order_Status
- Shipping_Address

Order_Item

Primary Key: OrderItem_ID

Foreign Keys: Order_ID, Book_ID

Attributes:

- OrderItem_ID (PK)

- Order_ID (FK)
- Book_ID (FK)
- Quantity
- Price

Payment

Primary Key: Payment_ID

Foreign Key: Order_ID

Attributes:

- Payment_ID (PK)
- Order_ID (FK)
- Payment_Method
- Payment_Date
- Amount
- Payment_Status
- Transaction_ID

Review

Primary Key: Review_ID

Foreign Keys: Customer_ID, Book_ID

Attributes:

- Review_ID (PK)
- Customer_ID (FK)
- Book_ID (FK)
- Rating
- Review_Text
- Review_Date

Reading_List (Shelf)

Primary Key: Shelf_ID

Foreign Keys: Customer_ID, Book_ID

Attributes:

- Shelf_ID (PK)
- Customer_ID (FK)
- Book_ID (FK)
- Shelf_Status
- Added_Date

Admin

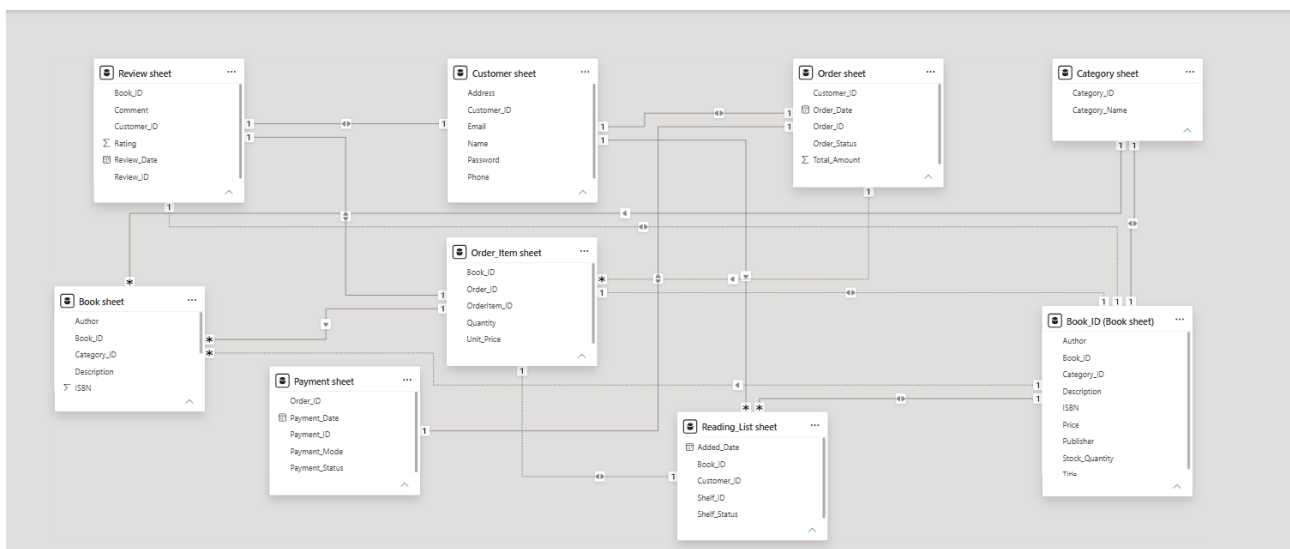
Primary Key: Admin_ID

Attributes:

- Admin_ID (PK)
- Admin_Name
- Email
- Password
- Role

2. Entities Relationship diagram:

The following figure represents the Entity Relationship Model of the Online Book Store project, designed using Power BI Model View. It illustrates the relationships between the entities based on their Primary Keys (PK) and Foreign Keys (FK), ensuring data integrity and efficient database management.



3. Relationships Between Entities

Relationship	Cardinality	Description
Customer → Order	One-to-Many (1:M)	One customer can place many orders, but each order belongs to one customer.
Customer → Review	One-to-Many (1:M)	A customer can write multiple reviews, but each review belongs to one customer.
Customer → Reading_List	One-to-Many (1:M)	A customer can add many books to their reading list/shelf.
Category → Book	One-to-Many (1:M)	One category can contain multiple books.
Book → Order_Item	One-to-Many (1:M)	One book can appear in many order items across different orders.
Book → Review	One-to-Many (1:M)	One book can receive many customer reviews.
Book → Reading_List	One-to-Many (1:M)	One book can be added to many customers' reading lists.
Order → Order_Item	One-to-Many (1:M)	One order contains multiple ordered items (books).
Order → Payment	One-to-One (1:1)	Every order has exactly one payment record.

3. Summary Keys

Primary Keys (PK)

- Customer_ID
- Category_ID
- Book_ID
- Order_ID

- OrderItem_ID
- Payment_ID
- Review_ID
- Shelf_ID
- Admin_ID

Foreign Keys (FK)

- Category_ID → Book
- Customer_ID → Order
- Order_ID → Order_Item
- Book_ID → Order_Item
- Order_ID → Payment
- Customer_ID → Review
- Book_ID → Review
- Customer_ID → Reading_List
- Book_ID → Reading_List

The Online Book Store – An E-Commerce Platform with Reading Tracker database consists of 9 entities that efficiently manage customers, books, categories, orders, order items, payments, reviews, reading lists, and administrators. The relationships between these entities ensure data consistency, eliminate redundancy, and support efficient order processing and inventory management. This relational database design provides a scalable and well-structured foundation for the successful implementation of the Online Book Store platform.

